5ARE0 Data Analysis and Learning Methods

Human Activity Recognition for Healthy Lifestyle Monitoring

Quartile 1 - 2025-2026

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1 Business & Objective Understanding

Wearable sensor technologies are able to continuously monitor physical activities of the user. Instead of self-reporting physical activity, daily physical activities can thus be detected and measured, which is great for promoting a healthy lifestyle. However, to be able to accurately measure the activity, devices first have to be able to distinguish between daily human activities.

Existing Human Activity Recognition (HAR) has been done using sensor-based classification to automatically detect and log daily human movements.

Therefore, the primary objective of this research is to develop and evaluate machine learning models that can automatically classify common daily activities using smartphone sensor data, with the goal of creating tools that support healthy lifestyle monitoring.

The five target activities that this research focuses on are sitting down, standing up, walking, running and climbing stairs.

Our research focuses on answering the folowing questions:

1. Which sensor-based features best discriminate between the five target activities (sitting down, standing up, walking, running, climbing stairs)?
2. How do supervised and unsupervised learning compare for this multi-class activity recognition problem?
3. Which machine learning model yields the highest classification accuracy when combined with appropriate feature extraction and selection techniques?
4. What are the practical limitations and considerations for deploying such a system in real-world health monitoring applications?

2 Data Understanding

2.1 Data Characteristics

The dataset contains time-series sensor data for five human activities: running, walking, sitting down, standing up, and stairs. In total, three participants collected the data, where each of us performed the activities in multiple sessions.

Using the sensors below:

- An accelerometer to measure the linear acceleration of the body in three axes (x, y, z) capturing movement intensity such as steps.
- A gravity vector sensor to detect the orientation and tilt of the body relative to the Earth's gravity.
- A gyroscopre to measures angular velocity or rotational motion around the three axes allowing twists and turns to be captured.

The sampling frequencies of all sensors were set to 100 Hz and standardization of the units has been turned on to allow for cross-sensor analysis.

2.2 Data Collection Methodology

To collect the data, a mobile application called "Sensor Logger" is used. This application allows you to record data using multiple sensors and allows you to export the data into .csv files per sensor.

All the participating researchers have collected data for each physical activity. The mobile phone has been stabilized around the left upper arm for every data recording.

2.3 Exploratory Data Analysis & Key Insights

In the initial phase of the exploratory analysis, the raw sensor signals were plotted over time for each activity. Some of the key observations made:

Walking and running show strong periodic oscillations in the accelerometer from rhythmic leg and arm motion, while the gravity sensor stays relatively stable, indicating an upright posture. In contrast, the stairs activity produces less regular accelerometer oscillations with greater vertical variation, the gyroscope shows irregular spikes from torso tilts, and gravity fluctuates mildly as the body leans forward during climbing. For sitting down and standing up, the accelerometer remains near zero except for short bursts of motion, the gravity sensor shows a clear slope change as the torso tilts between positions, and the gyroscope records brief but sharp rotational peaks during the transitions.

2.4 Data Quality Assessment & Limitations

Most recordings were smooth and consistent, but some showed small issues like sensor drift and minor noise. There were occasional slight timing differences between sensors at the start of recordings. Movements also varied between participants, such as walking speeds or stair heights, which added differences within the same activity. A few recordings had brief pauses or were shorter or longer than expected.

These limitations highlight the importance of preprocessing steps regarding the improvement of dataset quality.

3 Data Preparation

The raw sensor data collected for each activity required several preprocessing steps to ensure that it was clean, consistent and ready for modelling.

The steps followed included **loading and organizing** the sensor files, **cleaning and normalizing** timestamps, **trimming inactive periods** and **segmenting the recordings into overlapping time windows.** These steps reduce noise, align sensors, and increase the number of usable samples

3.1 Loading & Cleaning

All CSV files that contained the recordings of Accelerometer, Gyroscope, and Gravity sensors for each activity, were loaded into DataFrames. Files which contained information regarding the metadata were dropped, keeping only numeric features consisting of seconds elapsed and x, y, z values.

For activities like climbing stairs, sensor clocks started at different times, so the *seconds_elapsed* column was normalized by subtracting the first recorded timestamp. This ensured that Accelerometer, Gyroscope, and Gravity data were all aligned to start at 0 seconds, which is critical for trimming and windowing.

3.2 Activity Trimming

The raw data recordings contained noise at the very start and ends of the recording as the recordings captured the movements within the time frames where the phone was being placed and taken out of the arm band.

The trims were made by removing idle periods using RMS-based intersection method. The defined algorithms compute the RMS energy of the signal in sliding windows and marks the start and end where the RMS crosses certain treshold values. These treshold values were experimented specficially for each activity since each activity has a different RMS-base.

The decision to use an RMS-based intersection approach came after experimenting with several other methods. Initially, we tried comparing the average y-values of consecutive windows from the accelerometer data. Specifically, if the average of the next window was greater than or equal to twice the average of the current window (or vice versa), a cut was made. However, this approach proved unreliable. The y-values often fluctuated, causing cuts to occur at incorrect points and failing to capture the segments of interest. Additionally, variations in recording patterns across participants led the algorithm to perform well for some individuals but inaccurately for others.

3.2.1 Windowing with Overlap

After trimming the activities, the segments were divided into 3-second windows with 50% overlap. This approach was chosen to increase the number of samples and preserve temporal transitions between windows without losing important details.

Figure 1: Shaded regions depict the trimmed windows and the darker regions indicate the overlap of the trimmed windows

All windowed segments were combined into a single dataset where each window is labeled by its corresponding activity

3.3 Feature Engineering

After windowing, statistical and signal-based features were extracted from each window's x, y, and z axes for all sensors. These included mean, standard deviation, RMS energy, maximum and minimum values, and cross-axis correlations to capture both the magnitude and variation of movements. These features summarize the raw waveforms into more compact descriptors suitable for machine learning models.

3.4 Feature Selection

The selected features were then refined using both filter and wrapper methods (RFE, SFS) to retain the most informative ones for classification.

The filter methods used include a variance filter, to remove exactly-zero (dead) features, a correlation filter, to remove one of each highly correlated feature pair, and a mutual information filter that only kept the top-k features. K was chosen by plotting the change in accuracy of a baseline Logistic Regression model for different values of k. This resulted in k=30.

For every model, a different subset of the top-30 features was selected using a wrapper method. This wrapper method was either Recursive Feature Elimination (RFE) or forward Sequential Feature Selection (SFS), depending on whether the estimator exposes internal feature importances (only then faster methods, RFE, is possible).

4 Modeling

4.1 Supervised Learning Models

Three supervised learning algorithms were trained. These include a Logistic Regression (LR) model, a Decision Tree (DT), a Naive Bayes (NB) classifier, and a k-Nearest Neighbor (k-NN) algorithm.

Each of them was trained using grid-search for the optimization of the model's hyperparameters. This best model was determined using the cross-validation Cohen's Kappa score. All models were evaluated using Accuracy, Cohen's Kappa and F1-score, but Cohen's Kappa was used for model choices because it accounts for overall agreement across classes and also chance agreement.

Normalized confusion matrices were plotted for each model to be able to visualize and compare the classes which the different models have more trouble classifying. For each model, their corresponding metric for feature importance was used to visualize the features the models most rely on.

4.2 Unsupervised Learning Models

We evaluated K-Means, Fuzzy C-Means (FCM), and Gaussian Mixture Models (GMM) in a unified pipeline that standardizes features and uses grid-search with stratified cross-validation.

Models were selected by macro F1 after aligning cluster labels to ground truth; however, Cohen's Kappa is particularly well-suited to unsupervised evaluation because it adjusts for chance agreement and class imbalance, giving a conservative measure of cluster–class correspondence. For all unsupervised models, before training, a wrapper feature selector produced model-specific subsets, and we visualized errors with normalized confusion matrices to see which classes were most confused.

For K-Means, we fixed five clusters (matching the activities) and tuned the initialization restarts, iteration limit, and algorithm (Lloyd vs. Elkan) to trade off stability and speed under a spherical-cluster assumption. For FCM, we again used five clusters and tuned fuzziness (around 2.0), stopping tolerance, and iteration budget to control membership softness and convergence—helpful when activities overlap. For GMM, we set five components and varied covariance structure (full vs. diagonal), EM restarts, iteration limit, and covariance regularization to test whether flexible, anisotropic shapes fit better than spherical ones.

Cross-validation consistently favored FCM, with GMM close behind and K-Means trailing—reflecting their inductive biases: FCM's soft memberships capture overlap, GMM's flexible covariances handle correlated features, and K-Means' spherical assumption and init sensitivity yield weaker, more variable results. The wrapper-selected subsets likely amplified these advantages by preserving class-specific structure that FCM and GMM exploit more effectively than K-Means.

4.3 Comparison between methodology approaches

With labeled data available, supervised models are naturally advantaged: they train directly against the targets, tune hyperparameters on label-based metrics, and use every labeled sample to sharpen decision boundaries. Unsupervised clustering, by contrast, ignores labels during training, so model selection follows cluster geometry rather than classification error. This makes supervised training more sample-efficient and clearer for feature selection and validation, while unsupervised methods remain best for exploratory structure discovery or when labels are scarce or noisy.

5 Evaluation

5.1 Validation Strategy

In order to obtain a reliable estimate of generalization performance, a two-stage validation scheme was adopted.

5.2 Model Selection and Feature Training

All feature-selection procedures and hyperparameter searches were carried out exclusively on the training set. To mitigate variance introduced by class imbalance and participant-specific patterns, we used a 5-fold stratified cross-validation. Stratification ensured that each fold preserved the original class distribution, while random shuffling reduced bias from the recording order. At each CV iteration, the model was re-trained on 80% of the data and evaluated on the remaining 20%, and the process was repeated across folds to compute the mean and standard deviation of the chosen metrics.

For wrapper-based feature selection methods such as RFE and SFS, the 5-fold CV loop was executed at every step of feature subset selection. This ensured that subsets were chosen based on their cross-validated performance rather than on a single split, reducing the risk of overfitting to the training data.

5.3 Final Evaluation

The best hyper-parameters and feature subset for each algorithm were re-trained on the full training data and then evaluated on a test set that was never used during training or model selection. This selection between model selection folds and final test set aimed to provide an unbiased estimate of each model's generalization capability.

5.4 Performance metrics

The 3 reported metrics:

- Accuracy: The overall proportion of correctly classified windows.
- F1-score: Balances performace across classes (regardless of class frequency).
- Cohen's k (Kappa): Measures agreement between predictions and ground truth while correcting for chance.

Table 1: Model comparison results on the held-out test set.

Model	**Accuracy**	**Kappa**	**F1**
K-Nearest Neighbors	**0.95**	**0.94**	**0.95**
Decision Tree	0.90	0.88	0.90
Logistic Regression	0.90	0.88	0.90
Naïve Bayes	0.90	0.88	0.90
Gaussian Mixture (GMM)	0.89	0.86	0.88
Fuzzy C-Means	0.88	0.84	0.87
K-Means	0.64	0.55	0.54

Additionally plotted ormalized confusion matrices to visualize per-class performance and used PCA projections of the test set to highlight misclassified samples.

K-Nearest Neighbors achieved the best overall performance with 95% accuracy and k = 0.94, outperforming both other supervised models and all clustering-based approaches.

5.5 Discussion and Analysis

The overall results show that supervised models substantially outperformed unsupervised models for this activity recognition task. Among supervised classifiers, K-Nearest Neighbors (KNN), Decision Tree, Logistic Regression, and Naïve Bayes all achieved test accuracies around 0.90 or higher. KNN reaching the highest accuracy of 0.95 and the highest κ (0.94). In contrast, the unsupervised methods such as K-Means, Fuzzy C-Means, and Gaussian Mixture Models (GMM) lagged behind, with K-Means performing worst (0.64 accuracy and 0.55 κ).

This performance was expected since supervised models learn explicit decision boundaries using labeled data. This is well-suited for activity classes. Unsupervised models rely on clustering the feature space without labels, which is less effective when activity classes are not cleanly separable or when their distributions overlap.

The main limitation of supervised learning is the dependence on the labeled training data, meaning that the model's quality is tied to how representative and diverse the training set is.

Unsupervised models, while less accurate here, have the advantage of working without labels. They can be valuable in scenarios such as exploratory analysis of unannotated sensor streams, anomaly detection, or personalization when ground-truth activity labels are expensive or impractical to obtain. However, they struggle to capture complex timing between activities and usually require extra steps like post-processing.

Supervised learning is preferred for accurate classification when large labeled dataset exists. Unsupervised learning is better for data exploration, finding new behaviors, or when labeling data is impractical.

Although KNN had the highest accuracy (0.95), we used three metrics to pick the most reliable model. Relying only on accuracy alone was avoided since it's often unreliable in multi-class tasks, failing to show class balance or agreement beyond chance. KNN consistently scored highest on all three (Accuracy: 0.95, F1-macro: 0.95, : 0.94). Therefore, KNN is selected as the best-performing model overall.

KNN benefited from a combined feature set (time-domain statistics, frequency descriptors, and cross-axis correlations) which captured motion's magnitude, rhythm, and direction.

Feature selection via a filter–wrapper pipeline reduced redundancy and noise to improve generalization. The final, small feature subsets (5–15 features) were most effective when they included energy and correlation features for distinguishing dynamic activities.

Regarding misclassifications, they primarily occurred between activities with similar movement profiles, such as walking and slow walking. This is because their features overlapped. This difficulty was increased by inter-participant variability (different walking styles), which blurred class boundaries.

Overall, KNN was very accurate with a few errors (5%), which happened mostly during activity transitions. This highlights the need to add temporal context features to track changes better. Unsupervised models struggled more, often grouping unlike activities due to their low-features overlapping. To improve performance, future work should focus on adding temporal features and using hybrid semi-supervised models that can adapt to new, unlabeled data.

6 Deployment Considerations & Recommendations

The developed pipeline demonstrates a decent performance in controlled conditions where standardized data collection was used. For real-world use, the system would need to be deployed on mobile or wearable devices such as smart watches, which are capable of continuous sensor streaming and on-device inference.

In addition to algorithmic performance, deployment in the real world requires careful consideration of the hardware, software, and data acquisition pipeline. Devices must be capable of streaming sensors all the time, possess the processing power necessary for on-device inference, and have great connectivity for any optional cloud-based services. Data collection protocols must also allow for the variation in sensor placements and device types that will more generally be found outside laboratory environments.

This would require the following implementations:

- **Real-time data processing**: Windowing on the device and inference to recognize activities while the activity is taking place.
- **Resource constraints**: models must be lightweight to minimize battery consumption and memory usage.
- **Low-latency inference**: Especially important for real-time health or rehabilitation feedback.
- **Privacy and security**: Since accelerometer and gyroscope data can reveal sensitive behavioral patterns, strict local processing and data protection are crucial.

6.1 Limitations and Potential Improvements

A key limitation lies in the controlled way the dataset was collected. In our study, all participants wore the phone in a fixed orientation on the upper left arm, ensuring consistent sensor alignment across recordings. However, in real-world use, people often carry phones in pockets, bags, or hold them in their hands—or may not even use a phone at all, preferring other wearable devices such as smartwatches or dedicated activity trackers.

Such variations in placement, device type, and orientation can lead to significant differences in sensor readings, consequently reducing the model's overall performance.

For a robust deployment, future systems should implement sensor-placement-independent features or orientation-invariant preprocessing, and may benefit from multi-sensor fusion (e.g., combining wrist-worn devices with smartphones).

To address these challenges, future work should focus on:

- Designing **orientation-invariant features** or preprocessing steps that minimize the effect of device angle and placement.
- Extending the dataset to include **multiple device placements** (e.g., wrist, pocket, hip) and a more diverse participant pool.

- Exploring **multi-sensor fusion**, such as combining wrist-worn devices with smartphones to improve robustness.
- Collecting data under more realistic environmental conditions (e.g., outdoor terrain, uneven surfaces) to enhance generalization.

6.1.1 Recommendations for Future Work and Clinical Applications

While the current study demonstrates the feasibility of activity recognition using smart-phone based sensors there remain several points for improvement for future work. These improvements can aim to improve both technical robustness and clinical relevance.

As highlighted earlier, robustness to sensor placement and device variability remains a critical challenge. Future studies can aim to expand to include data from alternative placements such as the wrist, pocket, or ankle, and from other devices like smartwatches or dedicated activity trackers.

Similarly, there is always room for improvement through the use of a larger and more diverse dataset, with varied demographics, body types, and movement patterns. This is essential for improving model generalizability and clinical applicability.

Beyond these previously noted challenges, future research could focus on online and adaptive learning to personalize models over time, enhancing long-term monitoring in rehabilitation or elderly care.

Moreover, another important step will be the integration of the pipeline into clinical workflows. For example, doctors can use to track useful information for their patients during the post-surgery mobility tracking phase of their patients. Physiotherapists can also make use of this tool to track their patients improvements.

Finally, real-world validation outside controlled lab settings, along with strong attention to privacy, ethics, and usability, will be necessary to transition this proof-of-concept system into a trusted clinical and consumer-grade application.

7 Ethical and Privacy Considerations

1. All activities were performed by consenting adult participants with full understanding of the research purpose.
2. Participant safety was ensured at all times and trials were stopped if discomfort or injury risk arised
3. Data confidentiality: All sensor data remains anonymous and is used exclusively for this educational assignment
4. Informed participation: All group members understand data collection procedures and analysis goals
5. Data retention: Raw data will be deleted after course completion and final grade assignment

8 Technology Statement

During the preparation of this work, We used ChatGPT-5 in order to provide recommendations for code, justify unexpected results for personal understanding, and help with code and git errors. Parts 2-5 of the assignment were affected by AI tool usage. After using this tool/service, Liya, Berkan and Özde evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, Liya, Berkan and Özde take full responsibility for the content of their work.